Mercury had three main objectives: Place a manned spacecraft in orbital flight around the earth; Investigate man's performance capabilities and his ability to function in the environment of space; Recover the man and the spacecraft safely.

The first US spacecraft was a cone-shaped one-man capsule with a cylinder mounted on top. Two meters long, 1.9 meters in diameter, a 5.8 meter escape tower was fastened to the cylinder of the capsule. The blunt end was covered with an ablative heat shield to protect it against the 3000 degree heat of entry into the atmosphere. The capsule had very little room inside for the one person it held. In fact, the astronaut had to stay in his seat throughout the journey.

The Mercury program used two launch vehicles: A Redstone for the suborbital and an Atlas for the four orbital flights. Prior to the manned flights, unmanned tests of the booster and the capsule were made, carrying a chimpanzee.

Image Credit: Pyramid Design

A rendering of the Mercury spacecraft over the Sahara desert

The Mercury spacecraft was the first of its kind to have a failsafe mechanism to ensure that the pilot would not be harmed in case of a malfunction in the launch process. The Launch Escape System (LES) was a contraption of booster rockets attached to the spacecraft that would fire in case the launch rocket malfunctioned. The spacecraft would thus be boosted clear of the explosion and, after deploying a parachute, land safely.

Through a rigorous selection programme, the Mercury project narrowed down seven potential pilots from the US Air Force for the spacecraft. Known popularly as the Mercury 7, they were Scott Carpenter, Gordon Cooper, John Glenn, Gus Grissom, Wally Schirra, Alan Shepard and Deke Slayton.

While the Soviets preferred the term 'cosmonaut', they dubbed their men 'astronauts'. The term was a cross between 'aeronauts', as ballooning pioneers were called, and 'Argonauts', the legendary Greeks in search